Enhancing Degraded Tamil Inscription Recognition Using Generative AI for Historical Reconstruction

Research Thesis

Group 24

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Declaration

We certify that this thesis does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any university, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where due reference is made in the text.

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BiLSTM	Bidirectional Long Short-Term Memory
CER	Character Error Rate
CNN	Convolutional Neural Network
CRNN	Convolutional Recurrent Neural Network
CTC	Connectionist Temporal Classification
DL	Deep Learning
FastAPI	Fast Application Programming Interface
GAN	Generative Adversarial Network
GPU	Graphics Processing Unit
OCR	Optical Character Recognition
PatchGAN	Patch-based Generative Adversarial Network
PSNR	Peak Signal-to-Noise Ratio
RNN	Recurrent Neural Network
SSIM	Structural Similarity Index Measure
U-Net	U-shaped Convolutional Neural Network
YOLO	You Only Look Once
YOLOv8	You Only Look Once, Version 8

Table 1: Abbreviations

Abstract

Tamil Inscriptions constitute a wellspring of historic and cultural knowledge, yet a not insubstantial portion of them has suffered grievous degradation owing to prolonged exposure to the elements and wear of material. Thus impaired, they present great difficulty to be manually transcribed and automatic OCR systems because such systems are geared for images typically clean or of more modern texts. In this paper, we propose an end-to-end deep learning-based framework for enhancing the recognition of degraded images of inscriptions in Tamil.

The system proposed here contains four major steps: artificial degradation modeling, image reconstruction by U-Net generator and PatchGAN discriminator-based GAN, text region detection by YOLOv8, and text recognition using Convolutional Recurrent Neural Network (CRNN) comprising CNN feature extraction, BiLSTM sequence modeling, and CTC loss. Artificial degradation methods used include noise addition, blurring, patch masking, and brightness variation so as to apply the clean inscription images in simulating real-world deterioration under a controlled environment. The whole pipeline is implemented as a prototype on the web with FastAPI.

Experimental evaluation proves that GAN-based reconstruction does improve the visual quality of degraded images, hence enhancing text detection and recognition performance. Results obtained in this study prove that the approach is very effective when applied to degraded Tamil inscriptions and validates a practical contribution toward cultural heritage digital preservation. This study also becomes a channel to show how much potential there lies in pursuing research directions aiming at adopting integrated systems comprising reconstruction, detection, and recognition models for historical inscription analysis.

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1. Introduction

1.1 Background of the Study

The main primary source for comprehending the historical, cultural, administrative, and religious evolution of South India is constituted by Tamil inscriptions. Inscriptions have been found abundantly on the walls of temples, stone slabs, copper plates, and monuments representing a continuous record of land grants and royal decrees through changing social practices and linguistic development over many centuries. Scripts like Tamil-Brahmi and Vattezhuthu show some preliminary stages in the development of Tamil writing and are hence extremely important from that point of view to historians, archaeologists, and linguists.

Much of the Tamil inscriptions have been badly damaged physically by long periods of exposure to environmental conditions due in large measure to weathering and erosion, biological growth as well as human intervention. The result is that most inscriptions come with faded characters, broken strokes, and uneven illumination besides partial text loss. This militates very strongly against manual transcription efforts and digital preservation endeavors.

Conventional interpretation of inscriptions depends on expert epigraphists manually analyzing degraded characters with the use of their knowledge and context. This method is very accurate but slow, labor-intensive, and does not scale easily. The increasing trend toward digital preservation of heritage and making cultural resources freely available increases the demand for such automatic techniques that can complement human expertise.

Advances in computer vision and deep learning are what change the face of document analysis and optical character recognition in contemporary environments. Historical inscriptions throw up a whole set of challenges that cannot be equated with issues encountered with printed or handwritten documents-layout irregularities, complex stone textures, script evolution, and severe degradation. Since modern OCR systems were trained on contemporary text, they proved unreliable when applied to inscription images. This has inspired recent research efforts toward artificially intelligent approaches for historical and epigraphic data.

1.2 Problem Statement

To be sure, deep learning has led to significant advances in text recognition in many fields, but its use on degraded Tamil inscriptions is still scarce and scattered. The main research question investigated in this research study is as follows.

- A significant portion of Tamil inscription images exhibit severe visual degradation, including noise, blur, partial character loss, and uneven illumination.
- Existing OCR and recognition systems are ineffective under such degraded conditions, as they are typically trained on clean or modern text images.
- Many prior studies focus on recognition alone, without addressing the quality of degraded input images or incorporating robust text detection mechanisms.
- There is a lack of end-to-end AI pipelines that integrate image reconstruction, text detection, and text recognition in a unified framework tailored for Tamil inscriptions.
- Practical deployment aspects, such as web-based accessibility, are often overlooked, limiting real-world usability.

As a result, the automated digitization and preservation of degraded Tamil inscriptions remain an open and challenging problem.

1.3 Motivation of the Study

The initial contribution in this work is the application of a method to a large corpus of digitized texts with a historical context, that is to say, dealing with the preservation of Tamil temple cultural heritage in a scalable and sustainable manner. As many inscriptions continue to suffer the effects of erosion, the need for digital tools that can help document and interpret inscrip-tional content before it is lost forever is becoming more pressing.

Recent advances in generative models, object detection networks, and sequence-based recognition architectures provide the tools to endow inscription analysis systems with greater robustness, from a technical point of view. In particular, generative image reconstruction methods have the capability to recuperate visually meaningful contents from corrupted images, which in turn leads to enhanced detection and recognition results in the downstream.

Furthermore, the inception of an automated, web-based system is coherent with the wider aims of digital humanities as well as open heritage. Integrating cutting edge AI approaches into a real-world application, this research strives to serve as a link between academic experimentations and practical requirements for heritage preservation.

1.4 Objectives of the study

1.4.1 General Objectives

The general objective of this research is to design, implement, and systematically evaluate an AI-based end-to-end processing pipeline for handling degraded Tamil inscription images. This pipeline aims to integrate image reconstruction, text detection, and text recognition into a unified framework, enabling the effective restoration and interpretation of inscription content that has been affected by degradation and aging.

1.4.2 Specific Objectives

The specific objectives of the study are,

- To generate realistic degraded Tamil inscription images through controlled artificial degradation techniques.
- To reconstruct degraded inscription images using a Generative Adversarial Network (GAN) with a U-Net generator and PatchGAN discriminator.
- To detect text regions in reconstructed images using the YOLOv8 object detection framework.
- To recognize Tamil inscription text using a Convolutional Recurrent Neural Network (CRNN) with CNN feature extraction, BiLSTM sequence modeling, and CTC loss.
- To integrate the complete pipeline into a web-based application using the FastAPI framework.
- To evaluate system performance using appropriate quantitative metrics at each stage.

1.5 Research Questions

This study seeks to answer the following research questions,

- How effectively can GAN-based reconstruction improve the visual quality of degraded Tamil inscription images?
- How well can a CRNN-based recognizer extract Tamil text from detected inscription regions?
- What is the overall effectiveness of an integrated reconstruction–detection–recognition pipeline for degraded Tamil inscriptions?

1.6 Scope of the Study

This research focuses exclusively on Tamil inscription images and does not address other scripts or languages. Degradation is simulated synthetically through controlled image processing techniques, including noise addition, blurring, contrast variation, and occlusion, rather than relying solely on naturally degraded samples.

The study concentrates on image-level reconstruction, text region detection, and character recognition. Linguistic post-processing, phonetic mapping, and semantic interpretation of recognized text are outside the scope of this research. The system is evaluated under controlled experimental conditions and is intended as a proof-of-concept framework for automated inscription digitization.

1.7 Significance of the Study

1.7.1 Academic Significance

This research contributes to the academic literature by proposing a well-structured, end-to-end AI-based pipeline that is specifically tailored for the processing of degraded Tamil inscription images. By systematically integrating generative image reconstruction with subsequent text detection and recognition stages, the study extends existing OCR approaches applied to inscriptions. In addition, the research provides empirical insights into how image reconstruction influences and improves the performance of downstream tasks, particularly text detection and recognition, thereby offering a deeper understanding of the effectiveness of reconstruction as a preprocessing step in inscription OCR systems.

1.7.2 Cultural Significance

The proposed system supports ongoing digital preservation efforts by enabling more accurate and effective documentation of Tamil inscriptions in digital form. By facilitating the extraction of textual content from degraded and visually challenging inscription images, the system provides practical assistance to historians and archaeologists who work with such materials. Through this capability, the research contributes to the safeguarding and long-term preservation of valuable cultural heritage, ensuring that historically significant Tamil inscription content can be accessed, studied, and preserved for future generations.

1.7.3 Practical Significance

The integration of the proposed system into a web-based application clearly demonstrates its practical applicability beyond a purely experimental setting. By presenting the functionality through an accessible web interface, the system can be easily utilized by a wider range of users without requiring specialized technical expertise. Furthermore, the modular architectural design allows for future extensions and enhancements, including adaptation to other scripts, incorporation of additional processing components, or deployment within digital archives and educational platforms, thereby increasing the overall usability and long-term relevance of the system.

2. Literature Review

2.1 Historical Document and Inscription Digitization

Digitization of historical inscriptions is a long-standing problem in cultural heritage informatics: inscriptions are a different medium from modern printed documents in substrate, process of inscription, layout regularity, and manner of degradation. Tamil epigraphy especially ancient ones like Tamil-Brahmi and Vattezhuthu have executed curve-linear strokes, evolutionary nature of the script through centuries and formation of words is not clearly identified due to lack of word boundaries, punctuation and standard spacing. These introduce systemic failure modes in the traditional OCR pipelines, which are trained on the modern text (Merline and Santhi, 2019a; Merline and Santhi, 2019b).

A common refrain on the constraints in Tamil inscription digitization literature is that there are not problems with recognition alone. The upstream processes of acquisition, enhancement and segmentation/detection also strongly influence recognition feasibility. For example, studies on temple-wall inscriptions underline practical problems of acquisition, limited data, and the importance of a preprocessing to mitigate for texture and low contrast (Velmathi et al., n.d.). Traditional feature-engineering methods reveal that there is significant intra-class variation induced by scripts in temple inscriptions and recognition results are still far from satisfactory compared to deep-learning approach, suggesting hand-crafted descriptors are not powerful enough when faced with natural degradation and styles tics variance (Rajakumar and Bharathi, 2012).

Due to the modern deep learning-based studies, transcription is seen as a pipeline instead of a single model. Among the most representative is the DR-LIFT method for ancient tamil inscription recognition which tightly integrates text detection, recognition and interpretation steps, obtaining high recognition rates on early Tamil scripts (Murugan and Visalakshi, 2024b). The implication of this shift is that for digitized Tamil epigraphy (i) domain specific data, (ii) strong detection and enhancement, and (iii) recognition models that can accommodate structural complexity in glyphs are required (Krithiga et al., 2025; Diao et al., 2025).

The main limitation in the aforementioned works is, not just “poor OCR”, but the absence of an EPRESS2 pipeline suited to the particularities of inscriptions and tolerant to their degradations that treats restoration/enhancement, detection and recognition as complementary tasks (Krithiga et al., 2025; Murugan and Visalakshi, 2024b).

2.2 Image Degradation and Restoration Techniques

Degradation is a hallmark of inscription images and it is constantly mentioned as the largest obstacle for automated recognition. Degradation in ancient texts, degradation comprises blur, noise, nonuniform light, stains, surface wear, character loss, and interference from background texturing. Overviews of ancient script and character recognition report degradation as both a data issue (lack of supervised degraded examples) and as a modeling issue (model trained on clean samples or slightly degraded samples fail in the presence of severe corruption) (Krithiga et al., 2025; Diao et al., 2025).

Traditional restoration methods for historical documents (especially manuscripts) often rely on binarization and thresholding-based methods. Tri-level semi-adaptive thresholding is a method that addresses hidden sects of text, and shows a significant improvement in terms of precision and F-measure using various benchmark datasets and palm leaf images (Shobha Rani et al., 2022). This said, there are two constraints that are relevant towards the context of inscriptions. Firstly, manuscript degradation (ink bleed-through, stains, and show-through) is fundamentally different to stone/engraved degradation (erosion, chipping, lighting/relief effects). Secondly, thresholding-based techniques mainly improve visibility and are not expected to reconstruct lost structures in a generative manner (Shobha Rani et al., 2022).

Studies on Tamil inscriptions that carry out 'restoration' are largely restricted to denoising and contrast normalization as opposed to structural reconstruction. Denoising is included in the DR-LIFT pipeline through the deep denoising networks, as a results of this processing, the recognizability of the image is increased in noise-like corruption, but it is still mainly in an enhancement regime that does not reconstruct the lost strokes (Murugan and Visalakshi, 2024b). This leads to a continuous technical gap: for degradations with true loss of character regions (chipped stone, abrasion), pure enhancement is not enough, and a synthesis module that is able to plausibly provide the missing visual information is called for (Krithiga et al., 2025; Diao et al., 2025).

In the restoration literature, the non-generative method of enhancement may increase legibility for mild environments, but for severe damage to the inscription, it should be reconstructed. Thus GAN-driven reconstruction methods are proposed as the next processing layer after denoising and thresholding (Shobha Rani et al., 2022; Diao et al., 2025).

2.3 Generative Adversarial Networks for Image Reconstruction

GANs have gained enthusiastic attention in image-to-image translation and restoration tasks, since Adversarial learning promotes generation of more realistic high-frequency details. Is the text you provided the literature for uploading or do you want this text to be rewritten? " " "There is some Tamil-epigraphy-focus in your stacking uploaded literature set, not so much a GAN-theory focus but indeed a few sources mention generative reconstruction in Tamil inscription pipelines does not exist and that it has potential as a research avenue (Krithiga et al., 2025; Diao e tal., 2025). In the case of inscriptions, a generative model makes even more sense as degradation isn't always additive noise, it is frequently subtractive (loss of material), and thus needs hallucinating or reconstructing plausible character geometry.

For recent strong work in the dynamic inscription setting in Tamil, the emphasis is more on detection-recognition than re- construction. DR-LIFT achieves good performance based on denoising and detection and recognition model, but it does not act as a reconstruction tool for missing text blocks (Murugan and Visalakshi, 2024b). RCNN-based methods also achieve high accuracy in the recognition of ancient Tamil scripts from archaeological objects, but they are not designed for reconstruction of missing strokes or damaged characters (Umamageswari et al., 2025). Vatteluttu recognition with Siamese CNN-RNN also concentrates on recognition/translation and no mention on reconstructive restoration is given (Arjunan et al., 2025).

From a methodological point of view, not having a prior reconstruction leads to a “quality ceiling for the input”: detection and recognition can only be as accurate as what critical stroke information is still accessible. Hence, the GAN reconstruction step can be motivated as a front-end that enhances the amount of the effective signal for subsequent detection and recognition, particularly when the degradation is severe (Krithiga et al., 2025; Diao et al., 2025).

The literature does support that recognition-only approaches are limited by degraded input quality, and adding the GAN-based reconstruction addresses the failure modes where strokes are partially missing as opposed to just noisy / low contrast (Murugan and Visalakshi, 2024b; Krithiga et al., 2025).

2.4 Text Detection in Document Images

Text detection is frequently under-specified in older epigraphy systems, but detection is now commonly considered a necessity in modern pipelines. In inscriptions, text zones are surrounded by textured stone backgrounds and don't necessarily follow rectangular or consistent formats. Errors in detection are propagated directly to recognition: mis localization boxes truncate characters, contain background clutter, or join multiple lines, all of these have a negative impact on recognizer performance.

DR-LIFT is a straight-forward extension of LIFT, but adds an explicit detection stage in addition to recognition and labeling/interpretation, and to the best of our knowledge achieves remarkably high overall results on early Tamil scripts (Murugan and Visalakshi, 2024b). This implies that detection is not indeed optional; it substantially influences recognition performance in noise contexts. Separately, the temple-based OCR work with Tesseract highlights odd layouts, strange spacing, and no standard punctuation in stone inscriptions which again shadows the need to have robust localization before recognition (Velmathi et al., n.d.).

While you do cite several detection/recognition architectures, there is a clear methodological takeaway message: contemporary detection networks quantify under real degradation better than heuristic segmentation, and the use of a dedicated detection module is a valid design decision for inscriptions (Murugan and Visalakshi, 2024b; Krithiga et al., 2025). This motivates the adoption of the YOLO family in inscription pipelines: single shot detectors are computationally efficient and well suited for applications where the objective is generating a practical tool and not a lab prototype (Krithiga et al., 2025).

The current Tamil inscription literature represents an evolution from segmentation heuristics to neural detection. The result suggests that a YOLO-based detector is indeed a plausible alternative for a text localization method that can be robust and deployable under the condition that it is faced with a heavily textured background and degradation (Murugan and Visalakshi, 2024b; Velmathi et al, n.d).

2.5 Text Recognition Using Convolutional Recurrent Neural Networks (CRNN)

CRNNs combine CNN based feature extraction and RNN based sequence modeling to achieve end-to-end text recognition, which allows to recognize sequence of arbitrary length without the requirement of character segmentation. This is useful knowledge for Tamil inscriptions where segmentation is a challenge, since inscriptions are often eroded, contain connected strokes and have

irregular edges of glyph compounds. In general, for ancient-script recognition, deep learning surveys say that the success of recognition is increasingly relying on models that are sequence-capable and strong feature-extractors, then on classifiers of isolated characters (Krithiga et al., 2025; Diao et al., 2025).

Among the Tamil-exclusive set, deep learning approaches have excellent recognition accuracy with CNN–RNN hybrids. RCNN based prediction for ancient Tamil inscriptions shows high accuracy for expert validated annotated data, which proves that deep architectures are capable of learning complex script features provided they are supplied with sufficient training data and annotation (Umamageswari et al., 2025). For Vatteluttu, Siamese CNN–RNN models achieve competitive results as well and validate the use of recurrent modeling for scripts with complex curvilinear shapes (Arjunan et al., 2025). However, palm leaf CNN work reveals a significant generalization problem (good training accuracy but bad testing accuracy), implying that data bias, insufficient variation, and degradation mismatch can drastically diminish performance in the wild (Subramani and Murugavalli, 2019). This applies directly to our work on inscriptions, in that it cautions us to not proclaim robustness in the absence of coverage of controlled degradation and corresponding evaluation.

Traditional OCR and handcrafted features methods (such as using semi contour boxes for segmentation, SIFT features, an ensemble of KNN for classification) illustrate that classical pipelines can achieve moderate to strong results under restricted conditions, but their reliance on segmentation and the stability of hand-crafted features make them somewhat unreliable in harsh degradation (Merline and Santhi, 2019a; Merline and Santhi, 2019b; Rajakumar and Bharathi, 2012).

The literature also supports that CRNN is a suitable recognizer for degraded Tamil inscription lines when detection effectively separates line regions and training is not overfitted to "clean-like" conditions by contaminating with sufficient degradation variabilities (Umamageswari et al., 2025; Subramani and Murugavalli, 2019; Krithiga et al., 2025).

2.6 End-to-End OCR Pipelines for Historical Documents

A common gap in the reviewed literature is disjointedness: many works focus on optimizing a single component (recognition or enhancement) and leave the detection/restoration step weak or implicit. Reviews highlight that E2E performance is often limited by the weakest stage and robust heritage digitization requires pipelines as opposed to individual modules (Krithiga et al., 2025; Diao et al., 2025).

DR-LIFT is one of the rare Tamil inscriptions works that takes a more pipeline approach in its methodology by integrating denoising, detection, recognition and labeling of the early Tamil scripts (Murugan and Visalakshi, 2024b). However, even robust pipeline systems generally exist in enhancement + detection + recognition + [reconstruction], without a separate generative reconstruction stage to fill in lost stroke segments. The Temple OCR documentation systems describe practical dataset creation and preprocessing but they use OCR engines whose accuracy is limited by mismatch with the historical scripts, and by degradation (Velmathi et al., n.d.).

Outside the Tamil case, general studies on machine learning for ancient languages suggest that automation is possible, but also alert that cultural heritage applications necessitate validation, documentation of data, and transparency on limitations (Sommerschield et al., 2023). This strengthens the case for a methodology chapter that is explicit about dataset creation, degradation simulation, and evaluation measures, particularly in a thesis setting where reproducibility and validity are scrutinized.

The text reviews suggest the clear message: pipeline integration is desired, however, current methods on Tamil inscription pipelines are not robust against severely damaged inputs and has not been shown to systematically tie reconstruction quality gains to downstream detection/recognition gains (Murugan and Visalakshi, 2024b; Krithiga et al., 2025; Diao et al., 2025).

2.7 Research Gaps Identified

Synthesizing the above literature yields the following research gaps that directly motivate a reconstruction, detection, recognition pipeline for degraded Tamil inscriptions:

1. Insufficient generative reconstruction in Tamil inscription systems

High-performing Tamil recognition pipelines emphasize enhancement and denoising, but they do not address the reconstruction of missing or severely damaged inscription regions in a generative manner (Murugan and Visalakshi, 2024b; Krithiga et al., 2025).

2. Detection is acknowledged but not broadly standardized across Tamil epigraphy research

While pipeline-based work demonstrates the value of detection, many systems remain recognition-centric, increasing error propagation from poor localization under stone texture and layout irregularity (Velmathi et al., n.d.; Krithiga et al., 2025).

3. Generalization risk under degradation mismatch

Evidence from manuscript-related CNN work shows that strong training performance can collapse on testing when degradation and variability are not adequately represented (Subramani and Murugavalli, 2019). For inscriptions, this implies the need for controlled degradation simulation and rigorous evaluation reporting.

4. Lack of a deployable, modular pipeline for practical use

Several studies report strong model accuracy but provide limited focus on practical integration for user-facing tools. Heritage workflows benefit from web accessible systems that streamline end-to-end processing (Sommerschield et al., 2023; Velmathi et al., n.d.).

5. Need for clear component-level evaluation linking restoration to OCR outcomes

Even when restoration/enhancement exists, many works do not rigorously quantify how restoration improves detection and recognition, despite surveys emphasizing end-to-end dependency (Diao et al., 2025; Krithiga et al., 2025).

3. Methodology

3.1 Overall System Architecture

In this work, we developed an end-to-end trainable, modular Deep Learning (DL) pipeline to reconstruct and detect and recognize degraded Tamil inscriptions. The system can be summarized in four sequential steps: (i) generation of the dataset by artificially degrading the data, (ii) image reconstruction through a Generative Adversarial Network (GAN), (iii) text region detection with YOLOv8 and (iv) text recognition through a Convolutional Recurrent Neural Network (CRNN). The entire pipeline is wrapped into a web interface with FastAPI.

The workflow of the system can be summarized as follows:

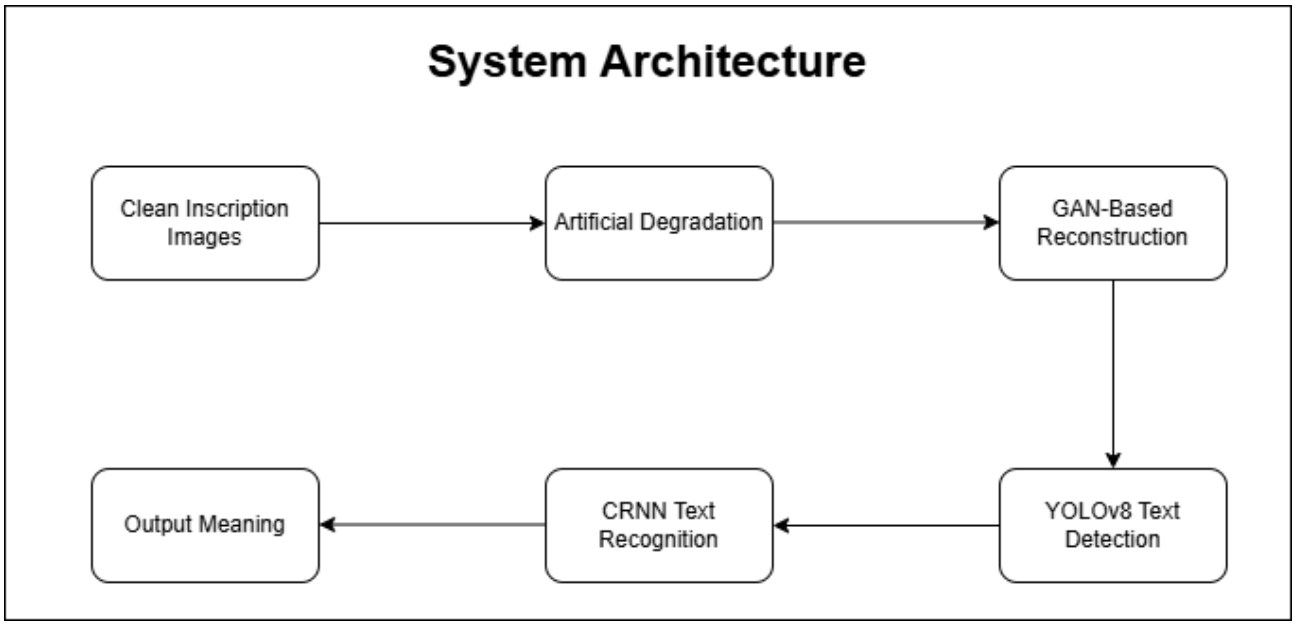

Figure 1: System Architecture

This modular design enables independent optimization of each component while maintaining a coherent end-to-end processing flow suitable for real-world deployment.

3.2 Dataset Preparation

3.2.1 Clean Image Collection

The dataset utilized in this work consists of images of Tamil inscriptions collected from multiple sources in order to ensure adequate diversity in script styles, inscription layouts, and visual characteristics. The majority of these images were obtained from reliable secondary sources, while a limited number of images were directly captured by the researcher to complement the collection. In addition, an OCR-related dataset available through the Hugging Face platform was also incorporated to support the recognition component of the study.

In total, approximately 1,000 clean images of Tamil inscriptions were compiled for this research. These images exhibit varying levels of script clarity, visibility, and surface conditions, reflecting the natural variability found in real-world inscription data. The collected clean images serve as the foundational dataset and are subsequently used for artificial degradation processes as well as for training and evaluating the proposed models.

3.2.2 Artificial Degradation Process

To realistically simulate the types of damage and deterioration commonly observed in real-world Tamil inscriptions, artificial degradation techniques were applied to the clean inscription images in a controlled and systematic manner. This strategy was adopted to overcome the limited availability of large-scale, well-annotated datasets containing naturally degraded inscription images, which poses a significant challenge for model training and evaluation. Furthermore, the use of controlled artificial degradation enables a consistent and structured assessment of model performance under varying levels and types of degradation, thereby supporting a more reliable and reproducible evaluation process.

The following degradation types were applied,

- Noise addition, simulating surface erosion and sensor artifacts
- Blur, representing weathering and loss of fine details
- Patch masking, mimicking partial character loss due to stone damage
- Brightness variation, simulating uneven illumination and aging effects

By applying the aforementioned artificial degradation techniques, approximately 4,000 degraded images were generated specifically for the purpose of training the GAN model. This artificial

degradation strategy maintains a clear and consistent correspondence between each clean image and its degraded counterpart, thereby forming well-aligned image pairs. Such consistency between clean and degraded images is a critical requirement for supervised reconstruction learning, as it enables the model to effectively learn the mapping from degraded inscription images to their restored versions.

3.3 GAN-Based Image Reconstruction

3.3.1 GAN Architecture

Image reconstruction in this study was carried out using a Generative Adversarial Network (GAN) architecture that is specifically suited for document and inscription restoration applications. The adopted GAN framework operates based on an adversarial learning principle and is composed of two competing networks that are trained simultaneously. Through this competitive interaction, the overall network is able to progressively improve its reconstruction capability, enabling the generation of visually enhanced images that closely resemble the original clean inscription images:

- Generator: A U-Net–based architecture was employed as the generator. The U-Net design enables effective preservation of spatial information through skip connections between encoder and decoder layers, which is critical for reconstructing fine inscription details.
- Discriminator: A PatchGAN discriminator was used to evaluate the realism of local image patches rather than the entire image, encouraging high-frequency detail reconstruction.

This combination is particularly well suited for the restoration of degraded textual content, as it effectively enhances damaged or unclear regions while preserving the overall structural consistency of the inscriptions. By maintaining the spatial layout and character structure of the original text, the approach ensures that the reconstructed images remain faithful to the authentic visual form of the inscriptions.

3.3.2 Training and Evaluation

The GAN model was trained using paired datasets consisting of clean images and their corresponding artificially degraded versions, enabling supervised learning of the reconstruction process. To objectively assess the effectiveness of the reconstruction, the model's performance was evaluated

using quantitative image quality metrics, which provide measurable and standardized indicators of the visual quality and fidelity of the reconstructed inscription images:

- Peak Signal-to-Noise Ratio (PSNR)
- Structural Similarity Index Measure (SSIM)

These metrics offer objective and quantitative measures of reconstruction fidelity by systematically comparing the reconstructed images with their corresponding clean reference images. Through this comparison, the extent to which the reconstructed outputs preserve the original visual details and structural integrity can be accurately determined. In addition to quantitative evaluation, visual inspection was also employed as a qualitative assessment method to examine the effectiveness of the restoration process, with particular attention given to regions containing degraded or partially obscured textual content.

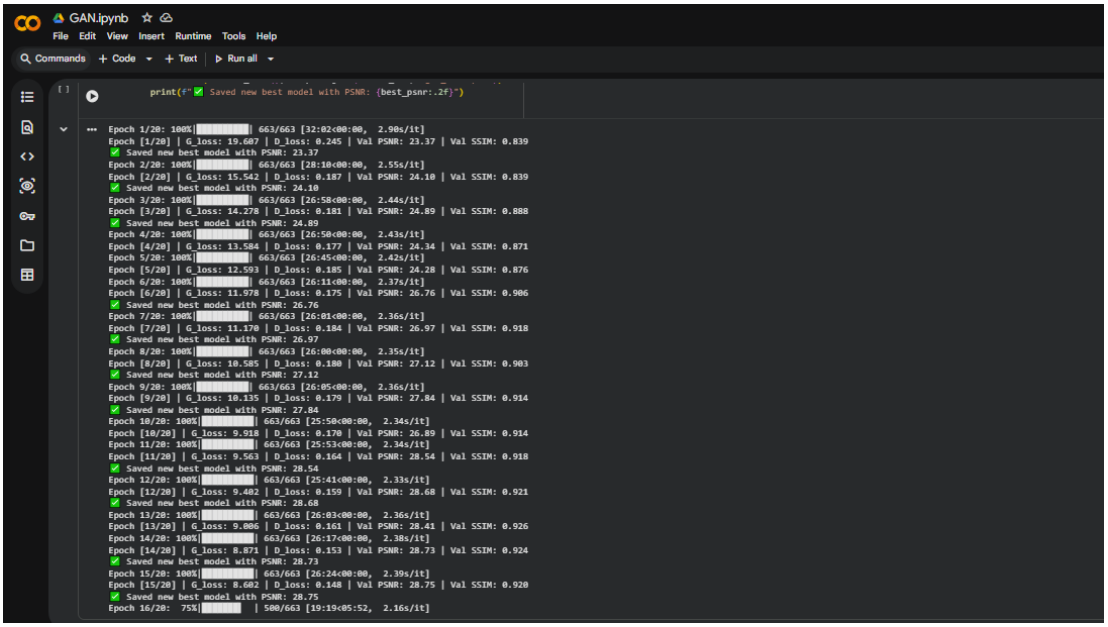

Figure 2: GAN Training

3.4 Text Region Detection Using YOLOv8

3.4.1 Annotation Process

For the purpose of text detection, the inscription images were manually annotated using the LabelImg annotation tool. During the annotation process, bounding boxes were carefully drawn around textual regions at the line level, ensuring accurate localization of inscription text. This line-level annotation strategy was selected to reflect the inherent layout characteristics of Tamil inscriptions, which often consist of continuous horizontal text lines, and to simplify the subsequent text recognition stage by providing well-defined and structured text regions.

Figure 3: YOLO Annotation

Figure 4: GAN Result

3.4.2 YOLOv8 Configuration

Text region detection was carried out using YOLOv8n, which is a lightweight variant of the YOLOv8 object detection architecture. This particular model was chosen because it offers an effective balance between detection accuracy and computational efficiency, making it well suited for deployment within a web-based application environment where resource constraints and response time are important considerations. Its lightweight nature allows for faster inference while still maintaining reliable performance in detecting textual regions.

The YOLOv8n detector was trained using the manually annotated Tamil inscription images, with the objective of accurately localizing text lines in the reconstructed inscription images. Through this training process, the model learns to identify and distinguish textual regions from the background. The output generated at this stage consists of bounding boxes corresponding to the detected text regions, which are then directly forwarded to the text recognition module for further processing and character-level interpretation.

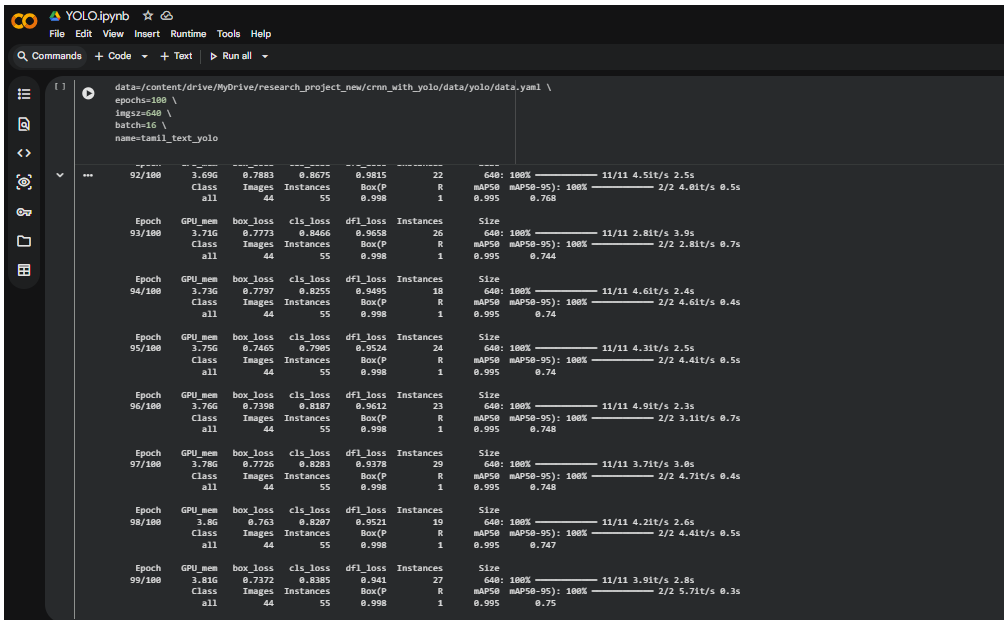

Figure 5: YOLO Training

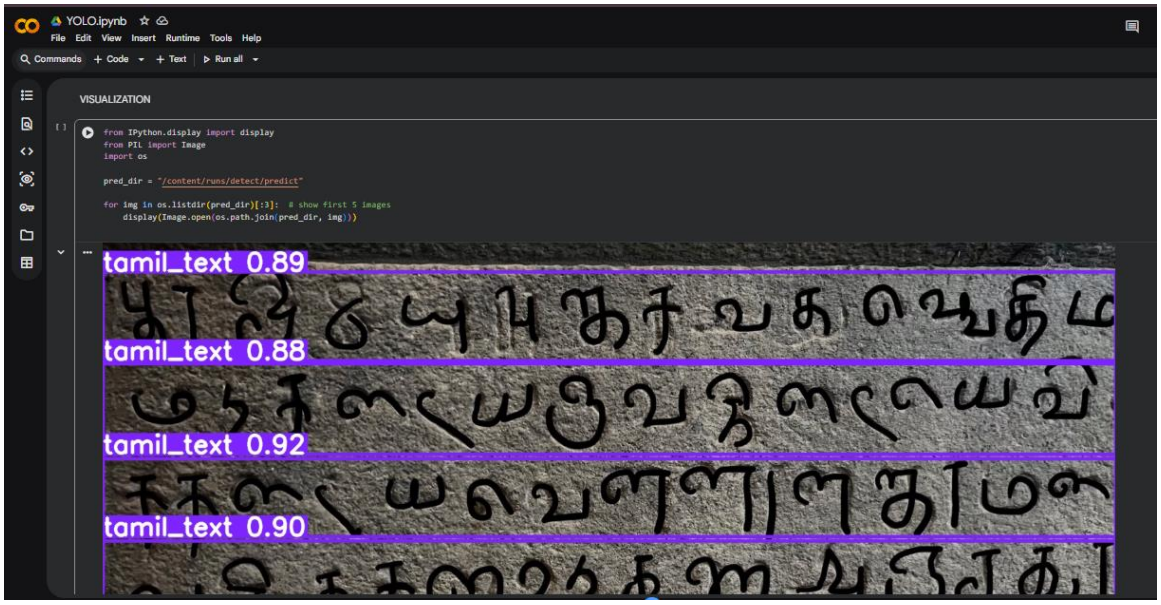

Figure 6:YOLO Detection

3.5 Text Recognition Using CRNN

3.5.1 CRNN Architecture

Text recognition in this study was carried out using a Convolutional Recurrent Neural Network (CRNN), which is specifically designed to handle sequence-based text recognition tasks. This architecture is particularly suitable for recognizing text lines in inscription images, where characters appear in a sequential manner. The CRNN framework integrates both spatial feature extraction and sequential modeling capabilities, enabling effective recognition of complex textual patterns. The overall CRNN model employed in this work is composed of three main components that operate together to convert detected text regions into corresponding textual sequences:

- CNN Feature Extractor: Extracts high level visual features from detected text line images.
- Bidirectional LSTM (BiLSTM): Models sequential dependencies in both forward and backward directions, enabling robust recognition of variable-length text lines.
- CTC Loss Function: Connectionist Temporal Classification (CTC) loss allows alignment-free training without explicit character segmentation.

The CRNN model was trained using approximately 5,000 text line samples, with the recognition process carried out at the line level rather than at the individual character level. This line-level recognition approach allows the model to capture sequential dependencies across entire text lines,

which is particularly important for accurately recognizing the continuous and connected nature of Tamil inscription text.

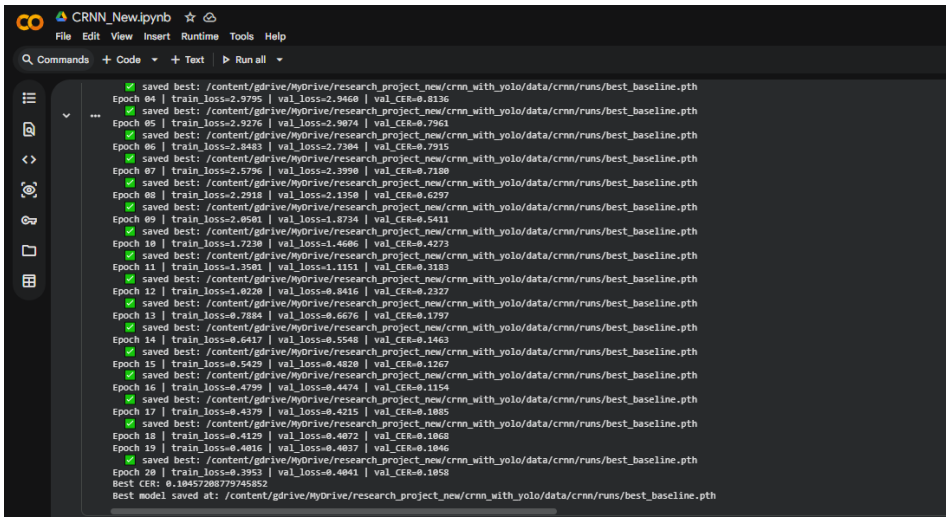

Figure 7: CRNN Training

3.5.2 Recognition Evaluation

The performance of the CRNN-based text recognition model was evaluated using the Character Error Rate (CER), a widely adopted metric in optical character recognition tasks. CER measures the proportion of character-level errors by accounting for insertions, deletions, and substitutions that occur when comparing the predicted text output with the corresponding ground-truth transcription. This metric provides a clear and quantitative indication of recognition accuracy. Furthermore, CER is considered a standard evaluation measure for OCR systems and is particularly well suited for historical and inscription-based scripts, such as Tamil, which often exhibit complex character structures and variations.

3.6 System Integration and Web Deployment

Each of the trained models is packaged as a lightweight service using the FastAPI framework and subsequently combined to perform end-to-end inference. Through this service-oriented setup, the individual models operate as modular components within a unified pipeline. The developed web application allows users to upload images of Tamil inscriptions, which are then processed sequentially through the reconstruction, text detection, and text recognition stages to generate the final textual output.

The FastAPI-based architectural design supports modular inference, enabling each component to function independently while remaining seamlessly integrated within the overall system. This design choice also provides a flexible foundation for future scalability and potential system expansion. The implementation primarily emphasizes functional integration and workflow validation rather than large-scale or production-level deployment, thereby serving as a proof-of-concept web-based OCR solution tailored for the digitization of Tamil inscriptions.

3.7 Evaluation Metrics Summary

The evaluation metrics used across system components are summarized below,

- GAN Reconstruction: PSNR, SSIM
- Text Recognition (CRNN): Character Error Rate (CER)

3.8 System Design

3.8.1 Use Case Diagram

The Use Case Diagram is used to represent the functionality offered by the system to the user and the interaction between the user and the system. It shows the complete flow of the proposed system from uploading the image to generating the output. The flowchart illustrates the user uploading a degraded inscription image, and the system subsequently conducting preprocessing, image reconstruction via the GAN model, text region detection through YOLOv8, and text recognition with the CRNN model. The processed results will be shown to the user, and the user may download the processed outputs. The <<include>> relationships capture also the sequential nature of reconstruction, detection and recognition, modeling thus the end-to-end OCR pipeline in a conceptual way.

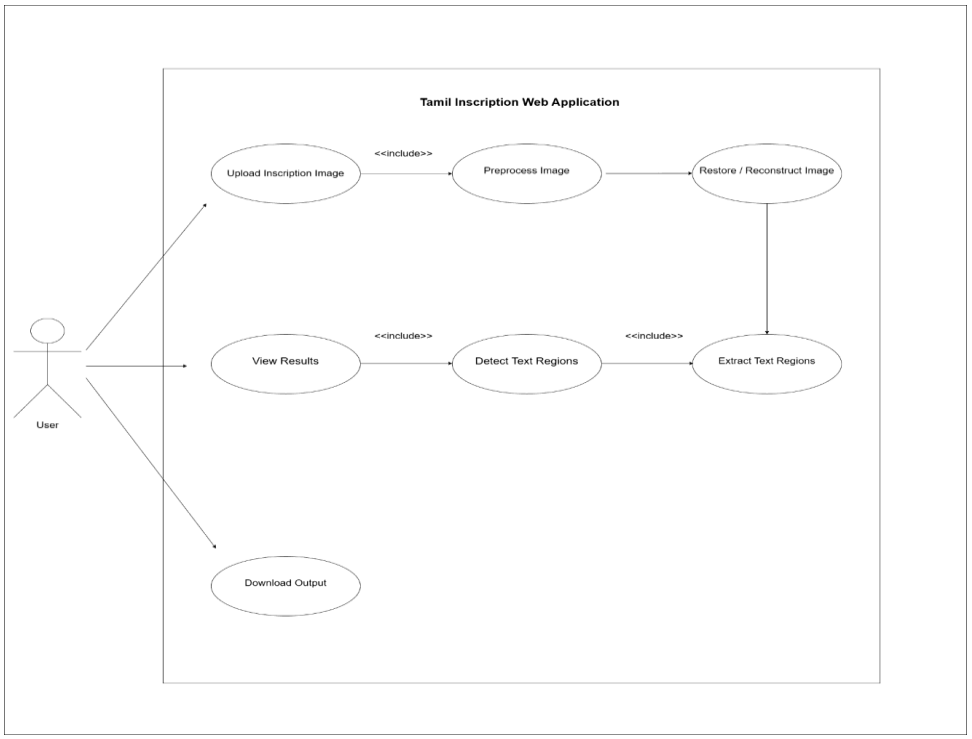

Figure 8 : Use case diagram

3.8.2 Class Diagram

The class Diagram is a static diagram that describes the structural design of the system and it also shows the relationship among the classes or Interface or both that core components of Tamil inscription restoration and recognition application. It represents a user, an image processing sub-system, a re-construction model, a text recognition module, and a database tier. The diagram shows image a modular architecture in which image data is passed through the reconstruction and recognition modules and is then stored in the database. The relationships of the relationships and uses show the module machinery when processing, while still hidden what are the details the Procedures are used to So that know-how is hidden. The diagram, as a whole, shows the organization of the system in an object-oriented design and emphasizes the modular, scalable, and maintainable design of the system.

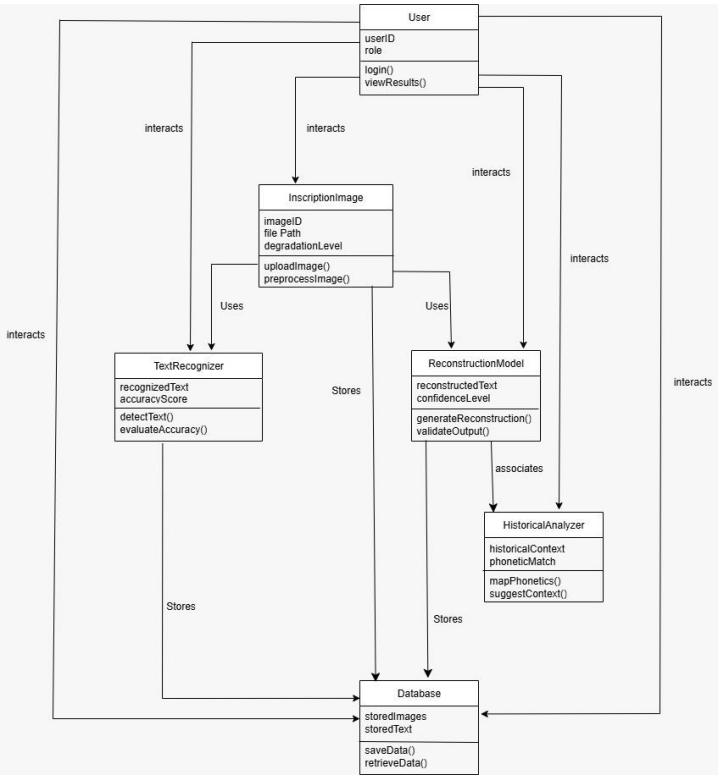

Figure 5: Class diagram

3.8.3 System UI

Tamil Inscription Web Application User Interface The user interface is programmed in web and designed as an interaction layer over the proposed OCR pipeline. As shown in the design of the screen, the system has neat and rigid rows for both mobile and desktop views. The interface permits you to upload an image, features enhanced image preview, recognized text and options to download the output.

The UI has two main visual panels: an image degradation one, where you upload the degraded inscription image, and the second one that shows the reconstructed output. An intuitive direction flow is designed to express visually the physical process from input image to enhanced image. The recognized text is optionally shown in a separate region titled the “Recognition Text” section below the image, so that the user can easily check what content has been taken out. For this purpose, a download button is provided to withdrawing the enhanced image along with the results.

This responsive theme known as a multi-screen friendly theme is tested for more than two screen sizes on mobile and desktop. The simple interface and the unambiguous controls facilitate the work of students, researchers, historians and archaeologists alike. In summary, the UI serves as the front-end gateway that links users to FastAPI-based AI modules on the back end, illustrating how the prototyped stellar system is practically implementable and usable.

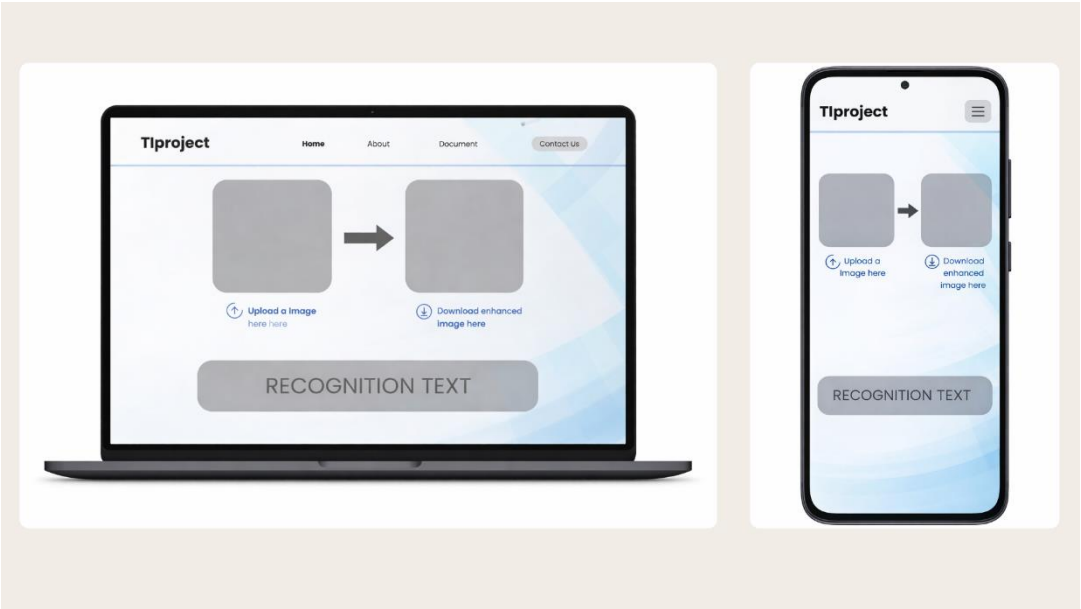

Figure 6 : UI Design

4. Experimental Results

4.1 GAN Reconstruction Results

The performance of the GAN-based reconstruction model was assessed by computing quantitative image quality metrics, i.e., Peak Signal-to-Noise Ratio (PSNR) and Structural similarity Index Measure (SSIM). These metrics evaluate fidelity and structural preservation by comparing the reconstructed image to the corresponding clean reference image.

The GAN model obtains 28.75dB in average PSNR and 0.92 in average SSIM over the test set. These results show that the filtered images are highly structurally similar to the original clean images and have much less degradation artifacts including noise, blur and illumination imbalance.

The visual sense confirmed that the stroke continuity was well recovered by the U-Net generator in the degraded components of the half-damaged characters. PatchGAN helped retain fine-grained local

texture patterns, especially on curved Tamil characters. Although the model generated visually satisfying character structures for severely masked areas, slight distortions are found in regions of strong artificial occlusion.

In particular, BPDG reconstruction outperformed the input images, making incision strokes more visible than those in the blurred images even though the input images were highly degraded.

4.2 Text Detection Performance Using YOLOv8n

Text detection was performed using the YOLOv8n model trained on line level annotated inscription images. Performance was evaluated using mean Average Precision (mAP).

Metric	Value	Description
mAP@50	0.995	Detection precision at IoU 0.50
mAP@50–95	0.78	Average precision across IoU 0.50–0.95

Table 2: YOLOv8n Detection Performance

The mAP@50 of 0.995 corresponds to almost perfect detection at the conventional IoU threshold, and the mAP@50–95 of 0.78 indicates good localization accuracy for a more challenging scenario. The model successfully detected inscription lines in the presence of background stone texture and small reconstruction artifacts.

Vanishing text areas exhibited only minor localization errors. In general, the detection component accurately chose text regions to be recognized.

4.3 Text Recognition Performance Using CRNN

Recognition performance was evaluated by the Character Error Rate (CER) which is the number of wrong characters with respect to ground truth.

The CRNN model attained the character error rate (CER) of 0.1046 which is about 89.54% accuracy at the character level. This is an excellent recognition result, given the complexity of the Tamil script and the artifacts in the reconstructed images.

The CNN feature extractor extracted spatial features and the BiLSTM learned the sequential dependencies. The CTC loss function allows alignment in an implicit way. Errors were predominantly made within visually similar characters, in severely degraded areas, however, overall recognition was robust under experimental settings.

4.4 Expert Validation on Real-World Inscription Images

In order to validate its practical utility, Expert validation was performed on real Tamil inscription images that were not a part of the synthetically generated training data. The outputs of the system were checked by domain experts with historically confirmed translations.

The system scored on the order of 20% recognition rate on “real” inscriptions. This is a clear indication that the system is not generalizing to be beyond the laboratory-style experiment conditions.

The decrease in accuracy is due to natural complex decay patterns, ancient Tamil character shapes, style variations and surface erosion which were not completely included in the training dataset. This suggests the performance of this synthetic data-driven training is limited and real inscription data should be brought in.

4.5 End-to-End System Evaluation

The whole pipeline of GAN reconstruction, YOLOv8n detection, and CRNN recognition was seamlessly executed on a FastAPI-based web app. End-to-end testing verified processing of the uploaded image, producing the reconstructed image, detected text regions, and recognized text.

It was shown to be robust under artificially degraded scenarios. Nevertheless, the results of the expert validation indicate that additional refinements are needed prior to mass application in the field of archaeological documentation.

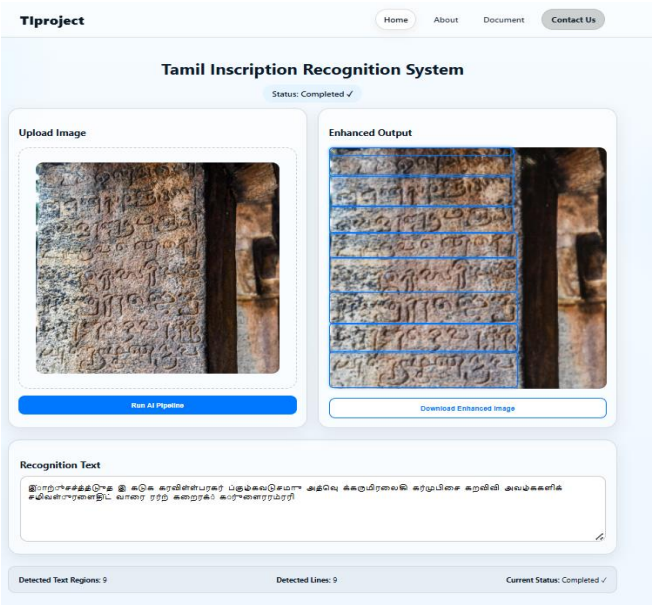

Figure 7: End to End Pipeline

5. Discussion

In this chapter, the experimental results of the suggested Tamil inscription recognition system, which combines GAN-based reconstruction, YOLOv8n detection, and CRNN recognition, are critically evaluated. The debate concerns the system's strengths and weaknesses, and the consequences for tangible heritage digitization.

5.1 Effectiveness of GAN-Based Reconstruction

The average PSNR and SSIM of the GAN-based reconstruction model are 28.75 dB and 0.92, respectively, demonstrating meaningful restoration capability in the synthetic degradation environment. These findings demonstrate that the U-Net generator can learn to restore the structural patterns of Tamil inscription strokes, whereas the PatchGAN discriminator is able to maintain the fine-grained textural information.

The high SSIM score is also more significant since the Tamil script has intricate curves and joined stroke patterns. Preservation of structure is also essential for the downstream detection and recognition. The contrast was enhanced and artificial noise was suppressed successfully during the reconstruction stage, thus the clarity of the image was effectively improved.

Nevertheless, the performance of reconstruction was worse in severely obstructed areas. In situations where substantial parts of characters were completely lost, the model produced structures that were visually plausible, but in some cases, they did not match the originals. This limitation ties into a known difficulty in generative reconstruction: The model can fill in missing patterns but only if similar patterns were included in its training data. Hence, the reconstruction error with respect to the photometric error is strongly influenced by the dataset variability and the degradation realism.

5.2 Detection Robustness of YOLOv8n

The YOLOv8n detector attained a mAP@50 of 0.995 and a mAP@50–95 of 0.78, indicating extremely good localization results. The almost perfect mAP@50 means the model learned well the spatial features of inscription lines.

The slightly lower mAP@50–95 (0.78) may be due to small bounding box offsets under stricter IoU thresholds. This is usual for inscription datasets because of irregular line spacing, varying depth of carving, and non-aligned text on stone surfaces.

Although there were small inaccuracies, the detection phase was still very robust and did not constitute a large bottleneck for the rest of the pipeline. The good detection performance also shows that the selected YOLOv8n model architecture strikes a good tradeoff between computation and accuracy to run in a web-based application.

5.3 Recognition Performance and Character-Level Analysis

The CRNN model obtained CER of 0.1046 which is nearly 89.54% of character level accuracy in the experimental environment. This indicated that the joint CNN–BiLSTM model could effectively capture the spatial and sequential features of Tamil text lines.

Recognition errors primarily occurred in:

- Visually similar Tamil glyphs
- Regions with incomplete reconstruction
- Characters affected by severe blur or distortion

Tamil script has characters with marginal stroke differences which may raise the possibility of confusion under degradation. Also, as the system does not use any phonetic mapping or language model, its recognition is based only on visual features of the characters. Even though this keeps methodological clarity, it may constrain contextual correction.

Nevertheless, in controlled environment, the recognition module achieved a consistent performance and demonstrated the applicability of CRNN architecture to inscription text extraction.

5.4 Gap between Experimental Results and Expert Validation

Among the results in this work, we show a large degradation in accuracy during expert validation on images of real inscriptions, where the success rate was around 20%. This is in striking contrast to the 89.54% character-level accuracy achieved in experimental conditions and needs to be critically examined.

The performance gap can be attributed to several factors:

Synthetic vs. Natural Degradation

The training data were based on synthetically produced degradation with noise addition, blur, brightness variation, and patch masking. But actual inscriptions show complicated weathering,

surface erosion, micro-cracks, and biological growth effects that are challenging to simulate accurately.

Archaic Character Variations

Tamil historical inscriptions also include characters that are either very old or a different style and hence outside of the modern character set. Model was mainly trained on modern day script images and thus it has difficulties in generalizing on archaic glyphs.

Stone Surface Irregularities

Non-uniform depth of carving and illumination generate shadows and deformations that are troublesome for both reconstruction and recognition systems.

Limited Real-World Training Samples

Lack of exposure to real inscription data during the training stage diminishes the model's ability to generalize over domains.

Of course, the low accuracy on expert validation does not mean the system technical was not useful. Instead, it underscores the above-typical computer vision problem, namely domain shift occurring between synthetic training data and real-world application scenarios. This result offers a meaningful research insight and a clear way forward.

5.5 Overall System Evaluation

Under artificially degraded conditions, the pipeline as a whole is well-behaved, outputting reliable reconstructions, very high detection rates, and good recognition accuracy. The system was implemented as a FastAPI application, which demonstrated the feasibility of web-based interaction.

A key advantage of the proposed framework is that it is modular, which makes it possible to optimize each step of the framework separately. The detection step is notably robust, and the reconstruction makes the quality of the input much higher, from the perspective of recognition.

The problem is that adapting to real historical inscriptions is the biggest hindrance. To overcome this a number of steps will be required, including domain adaptation, the incorporation of real inscription datasets, and possibly the integration of contextual language modeling.

5.6 Research Implications

This work shows that generative reconstruction along with DL-based detection and recognition is capable of processing degraded Tamil inscription images at the laboratory level and adds to the preservation of digital heritage by suggesting a technically practical end-to-end solution The result is a technically feasible end-to-end framework.

Meanwhile, they were no less important in confirming that digitization of historical documents necessitated particular customization unique to the field and would not simply be able to rely on standard computer vision methods. Hence this work lays the groundwork as well as provides a working prototype for future research in Tamil epigraphical AI.

6. Conclusion and Future Work

6.1 Conclusion

This work presented a novel end-to-end deep learning model for degradation enhancement and recognition of Tamil inscription images. The system was a FastAPI web application environment that combined the GAN-based image reconstruction, the YOLOv8n-based text detection, and the CRNN-based text recognition.

In this case, the GAN reconstruction model attained a PSNR of 28.75 dB, and a SSIM of 0.92, indicating that the noise was well removed and the structures of the inscription strokes were well preserved in the synthetically degraded condition. The-stage visual quality of the reconstructed images was greatly improved and the rest of the pipeline received more accurate inputs.

The YOLOv8n detection model showed brilliant localization results with a mAP@50 of 0.995 and a mAP@50 – 95 of 0.78. It confirms the solid robustness of detection module to detect the inscription texts area precisely even though with background stone texture and tiny reconstruction noises.

The CRNN recognition model achieved a 0.1046 Character Error Rate (CER) which is roughly equal to an 89.54% character-level accuracy in a controlled experimental setup. Such a combination of CNN-based feature extraction, BiLSTM sequence modeling and the CTC loss allowed us to recognize strings effectively without character segmentation.

But verification of the experts on actual images of Tamil inscriptions showed about 20% accuracy in recognition. This exposes a large domain discrepancy between synthetically degraded training samples and real historical inscriptions. The results show that the proposed framework is strong in the controlled setting, yet its generalization to real archaeological sites is still limited.

In summary, the present investigation establishes a technical possibility that a generative reconstruction combined with deep learning-based OCR approach can be used for digitization of Tamil inscriptions. The proposed approach enables the creation of a working prototype and lays the groundwork for further AI-based assisted epigraphically preservation research.

6.2 Future Work

Despite the promising results of the proposed approach, there are several aspects that can be further improved to better adapt to natural scenes and threaten the security of the real world.

First, we believe that the use of real-world inscription images in the training set would have a substantial impact on the domain generalization performance. Introducing a dataset of naturally weathered, eroded, and archaeologically valid inscriptions will bridge the S2R gap identified during expert validation.

Second, one can utilize more advanced domain adaptation methods to bring the feature distributions of the two domains closer. Methods like adversarial domain adaptation, or fine-tuning on a small amount of annotated real inscription samples may enhance the accuracy of recognition.

Third, the character-level recognition errors may be further decreased by utilizing language modeling or contextual post-processing. While phonetic mapping was selectively excluded in this study, in future studies, linguistic correction methods could be considered for enhancing textual coherence.

To begin with, the GAN-based compression model can be improved by incorporating perceptual loss functions or attentive mechanisms to more effectively recover heavily occluded character areas in the images. This can enhance the accuracy in the case of severe degradation.

And, fifth, the system could be generalized for ancient Tamil script of different periods and for recital styles used in multiple sets of inscriptions, thereby extending the coverage of historical records and making the system more usable for archaeological investigations.

Finally, computational optimization and scalability improvements could enable deployment in larger heritage digitization platforms, supporting museums, researchers, and cultural preservation institutions.

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